

Modular Thermality of the Celestial Spectral Weight: Planck Form, Matsubara Poles, and the Moment Family

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Abstract

The weight $P(\lambda) = \pi\lambda / \sinh(\pi\lambda)$ governs boost integrals throughout celestial holography. A companion paper derives it from two-particle phase space, where it appears as $\Gamma(1 + i\lambda)\Gamma(1 - i\lambda)$ on the shadow locus. This paper collects its thermal and modular structure, with every identity recomputed independently to at least 24 digits.

The weight takes a Planck form, $P = 2\pi\lambda[n_B(\pi\lambda) - n_B(2\pi\lambda)]$, in which the zero-point energies of the two Bose terms cancel exactly. Read through the Bisognano–Wichmann theorem, it is the partition function of one oscillator at the modular temperature $\beta = 2\pi$, weighted by boost frequency. Five further descriptions of the same function are given: squared Gamma modulus on the critical line, characteristic function of the logistic distribution, Weierstrass product over the Laplace spectrum of the thermal circle $\mathbb{R}/2\pi\mathbb{Z}$, meromorphic function whose poles sit exactly at the bosonic Matsubara points $\lambda = in$ with residues $i(-1)^n n$, and generator of cumulants that are even zeta values. The Mellin moments form a one-parameter family,

$$\mathbf{m}_s = \frac{1}{2\pi} \int_0^\infty \lambda^{s-1} P(\lambda) d\lambda = \pi^{-(s+1)} (1 - 2^{-(s+1)}) \Gamma(s+1) \zeta(s+1),$$

verified at five values of s ; the members $\mathbf{m}_1 = 1/8$ and $\mathbf{m}_3 = 1/16$ are exact, and the family is the structural twin of the Stefan–Boltzmann integral, with $\zeta(s+1)$ where the cavity law has $\zeta(4)$.

The physical thesis of the superseded draft is withdrawn. The claim that cavity radiation and the ultraviolet behaviour of quantum gravity are one disease with one cure rested on a companion claim, since retracted, that P replaces the Feynman loop measure. It does not. What remains is a precise thermal reading of a function that arises from phase space, which is worth stating on its own terms and is not worth more than that.

Keywords: celestial holography, modular theory, Bisognano–Wichmann, Matsubara frequencies, Dirichlet lambda function.

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1 What the weight is

The superseded draft opened by positing $P(\lambda)$ as the measure that cures loop divergences, then sought a thermal reading of it. The present order is the reverse, and it is the order the evidence supports.

Theorem 1.1 (Origin, from [1]). *The Mellin image of two-particle massless phase space in celestial coordinates is*

$$\int d\Pi_2 \omega_5^{\Delta_5-1} \omega_6^{\Delta_6-1} = \frac{1}{8\pi} \left(\frac{M}{2}\right)^{\Delta_5+\Delta_6-2} \frac{\Gamma(\Delta_5)\Gamma(\Delta_6)}{\Gamma(\Delta_5+\Delta_6)}, \quad (1)$$

and on the shadow locus $\Delta_5 + \Delta_6 = 2$ with $\Delta_5 = 1 + i\lambda$ the Gamma ratio equals

$$\Gamma(1+i\lambda)\Gamma(1-i\lambda) = \frac{\pi\lambda}{\sinh(\pi\lambda)} =: P(\lambda). \quad (2)$$

So P is a phase-space density of a unitarity cut. Everything below describes that density; nothing below licenses replacing a measure anywhere.

2 Planck form and the modular temperature

Theorem 2.1 (Planck form, with cancelling zero-point energy). *Write $n_B(y) = (e^y - 1)^{-1}$ and $E(x) = \frac{x}{2} \coth \frac{x}{2}$. For all $\lambda > 0$,*

$$P(\lambda) = 2\pi\lambda [n_B(\pi\lambda) - n_B(2\pi\lambda)] = 2\pi\lambda \sum_{n \geq 0} e^{-2\pi\lambda(n+\frac{1}{2})}, \quad 2E(\pi\lambda) - E(2\pi\lambda) = P(\lambda). \quad (3)$$

The zero-point contributions cancel identically: $2 \cdot \frac{\pi\lambda}{2} - \frac{2\pi\lambda}{2} = 0$.

Proof. Both reduce to $\coth y - \coth 2y = 1/\sinh 2y$. Verified at $\lambda = 0.4, 1.3, 3.1$ (first form) and $\lambda = 0.9, 2.6$ (second) to relative error below 3.3×10^{-39} . $\square$

Theorem 2.2 (Gibbs reading at $\beta = 2\pi$). *Let K generate the Lorentz boost preserving a Rindler wedge whose edge is the celestial sphere. By the Bisognano–Wichmann theorem [4, 5], the Minkowski vacuum restricted to the wedge algebra is a KMS state at inverse temperature $\beta = 2\pi$ in boost time. The middle expression of (3) is the partition function of a single oscillator with level spacing $2\pi\lambda$ in that state, multiplied by the boost frequency $2\pi\lambda$.*

The cancellation in (3) deserves a word. A single Bose mode carries $\frac{x}{2}$ of zero-point energy, which is what makes the naive sum over modes diverge. The combination appearing in P is a difference of two such modes at spacings $\pi\lambda$ and $2\pi\lambda$, and the zero-point pieces cancel between them. The finiteness of P at large λ therefore has a clean thermodynamic reading. It is a statement about this function, and it says nothing about loop integration; see Sec. 5.

3 Five further faces of the same function

Proposition 3.1 (Equivalent descriptions). *Let $P(\lambda) = \pi\lambda/\sinh(\pi\lambda)$.*

- (i) Gamma modulus on the critical line. $P(\lambda) = \Gamma(1 + i\lambda)\Gamma(1 - i\lambda) = |\Gamma(2s)|^2$ at $s = \frac{1}{2} + \frac{i\lambda}{2}$. Checked to 7×10^{-41} .
- (ii) Logistic characteristic function. $P(\lambda) = \int_{\mathbb{R}} e^{i\lambda x} dx / (4 \cosh^2(x/2))$, so P is the characteristic function of the logistic distribution, equivalently the Fourier transform of the reflectionless Pöschl–Teller ground-state density; the transform in the other direction is $\hat{P}(k) = \frac{\pi}{2} \operatorname{sech}^2(k/2)$. Checked to 1.5×10^{-31} .
- (iii) Functional determinant. $P(\lambda)^{-1} = \prod_{n \geq 1} (1 + \lambda^2/n^2)$, a Weierstrass product over the Laplace spectrum of the thermal circle $\mathbb{R}/2\pi\mathbb{Z}$. Checked at $\lambda = 0.8, 2.5$ to 3.1×10^{-31} .
- (iv) Matsubara poles. The continuation of P is meromorphic with simple poles exactly at $\lambda = in$, $n \in \mathbb{Z} \setminus \{0\}$, and $\operatorname{Res}_{\lambda=in} P(\lambda) = i(-1)^n n$. Checked at $n = 1, 2, 3$ to 10^{-12} by contour evaluation. At $\beta = 2\pi$ the bosonic Matsubara frequencies are $\omega_n = 2\pi n/\beta = n$, so the poles sit at the Matsubara points of the modular temperature.
- (v) Cumulants are even zeta values. For $|\lambda| < 1$, $\log P(\lambda) = -\sum_{k \geq 1} \frac{(-1)^{k+1} \zeta(2k)}{k} \lambda^{2k}$.

Item (iv) is the sharpest of the five for physical purposes. A weight whose poles land on Matsubara frequencies at $\beta = 2\pi$ behaves, under contour deformation, the way a thermal propagator behaves at that temperature. That is a structural fact about boost integrals in flat space, and it is the reason the same function keeps appearing in otherwise unrelated celestial calculations.

4 The moment family

Theorem 4.1 (Moment family). *For every real $s > 0$,*

$$\mathbf{m}_s := \frac{1}{2\pi} \int_0^\infty \lambda^{s-1} P(\lambda) d\lambda = \pi^{-(s+1)} (1 - 2^{-(s+1)}) \Gamma(s+1) \zeta(s+1). \quad (4)$$

In particular $\mathbf{m}_1 = \frac{1}{8}$ and $\mathbf{m}_3 = \frac{1}{16}$.

Proof. Insert the exponential series of (3). Termwise integration is justified by monotone convergence, since every term is positive:

$$\mathbf{m}_s = \int_0^\infty \lambda^s \sum_{k \geq 0} e^{-(2k+1)\pi\lambda} d\lambda = \frac{\Gamma(s+1)}{\pi^{s+1}} \sum_{k \geq 0} (2k+1)^{-(s+1)}, \quad (5)$$

and the odd-integer sum is the Dirichlet lambda function $(1 - 2^{-w})\zeta(w)$ at $w = s+1$. Verified independently at $s = 0.7, 1, 2, 3, 4.5$ with relative error at most 2.5×10^{-31} ; the values $\mathbf{m}_1 = 0.125$ and $\mathbf{m}_3 = 0.0625$ came out exact at working precision. $\square$

Remark 4.2 (The analogy, sized correctly). The Stefan–Boltzmann law arises from $\int_0^\infty x^3/(e^x - 1) dx = \Gamma(4)\zeta(4) = \pi^4/15$. Equation (4) has the same shape, with the Dirichlet lambda function in place of zeta, because the Bose difference in (3) restricts the sum to odd integers. The two formulas are cousins in the sense that both are Mellin transforms of Bose-type kernels. The superseded draft went further and treated the resemblance as evidence that one law cures the pathology of the other. It is not evidence of that, and the inference is withdrawn.

Remark 4.3 (On $\mathfrak{m}_1 = \mathcal{M}_1 = 1/8$). The ladder moment $\mathcal{M}_1$ of [2] is the $s = 1$ member of this family, so those two $1/8$ s agree by construction. A third $1/8$, appearing in the digamma moment of [3], comes from an integral over the full line and is a separate evaluation. The agreement of that one with these remains unexplained, and should be read as a coincidence until someone derives it.

5 Retractions

The following are withdrawn from `blackbody_law_qg_dtoupin_v1`, and are listed by name so that readers holding that draft can mark it.

1. **“The ultraviolet catastrophe of cavity radiation and the ultraviolet divergence of quantum gravity are the same disease, and admit the same cure.”** Withdrawn. The claim depended on P acting as a replacement for the Feynman loop measure, which is retracted in [2]. The companion derivation [1] reconstructs the standard loop integral from its cut exactly, divergences included.
2. **“ P renders the celestial loop expansion finite at every order.”** Withdrawn. The bound $\mathcal{M}_L \leq (1/8)^L$ is true of the convolution integrals in [2] and carries no implication for amplitudes.
3. **“ $\mathfrak{m}_1 = 1/8$ is the gravitational Stefan–Boltzmann constant.”** The value is correct, and Theorem 4.1 proves it. The title attached to it presumes the loop-measure reading, so the name is withdrawn while the number stands.
4. **Claims inherited from the companion drafts.** The description of P as the Plancherel density of $\mathrm{SL}(2, \mathbb{C})$ is false; the density for the scalar principal series grows like λ^2 . The fifteen-digit box verification cited in support was circular, since the routine it used reparameterizes a Feynman integral and compares it with itself.

6 Status of each statement

Statement	Status	Check
P from phase space	proved in [1]	10^{-41}
Planck form; zero-point cancellation	proved	3.3×10^{-39}
Gibbs reading at $\beta = 2\pi$	exact identification	
Gamma modulus on critical line	proved	7×10^{-41}
Logistic characteristic function	proved	1.5×10^{-31}
Weierstrass product	proved	3.1×10^{-31}
Matsubara poles, residues $i(-1)^n n$	proved	10^{-12}
Cumulants are even zeta values	classical	
Moment family (4)	proved	2.5×10^{-31}
$\mathfrak{m}_1 = 1/8$, $\mathfrak{m}_3 = 1/16$	proved	exact
Cavity radiation as a cure for QG	withdrawn	
Finiteness of the loop expansion	withdrawn	

All numerical checks are reproduced by `verify_blocks_v2.py`, archived with this paper.

References

- [1] D. Toupin, “Loop amplitudes from unitarity cuts in celestial Mellin space: a complete derivation for the scalar box,” v19, 2026.
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- [5] J. Bisognano, E. Wichmann, J. Math. Phys. 17 (1976) 303.